Workstation Critical Care/Neonatal Care; EvitaV300
Option PS500

PS500 supplies workstation CC&NC components and EvitaV300 direct with 24V

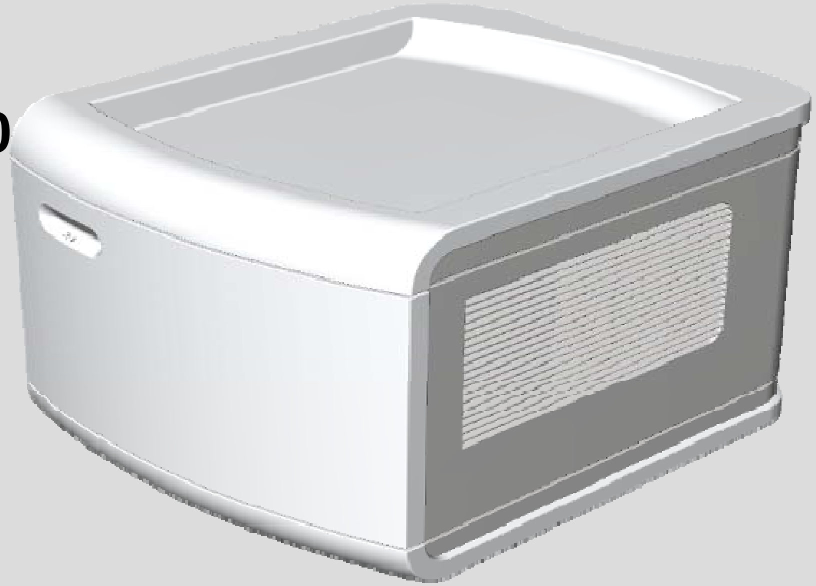
- **C500/V500 & C500/VN500 & C300/EvitaV300**
- **GS500**

- **Battery backup of workplace/EvitaV300**

  - 240* minutes (w/o GS500)

  - 120* minutes (with GS500- DC)

- **see IfU V500/VN500/V300 for more info**
- **available since April 2011**
- **NeBaCo since 01/2017**

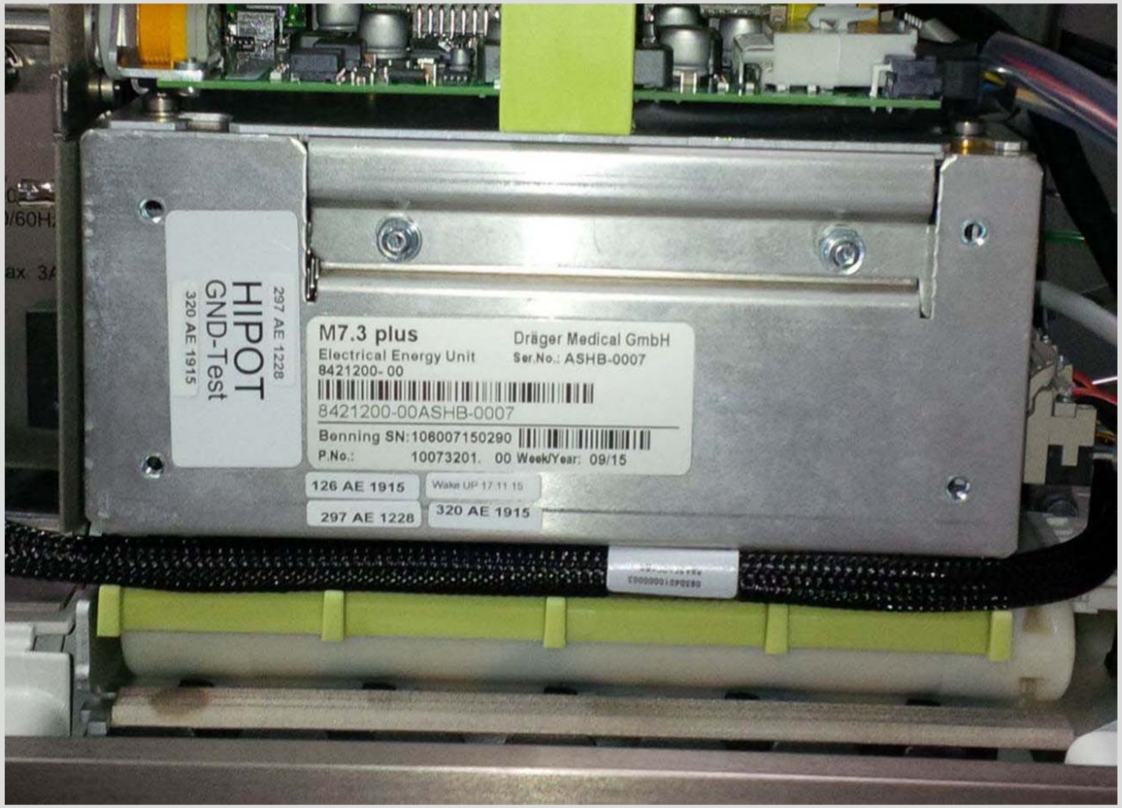


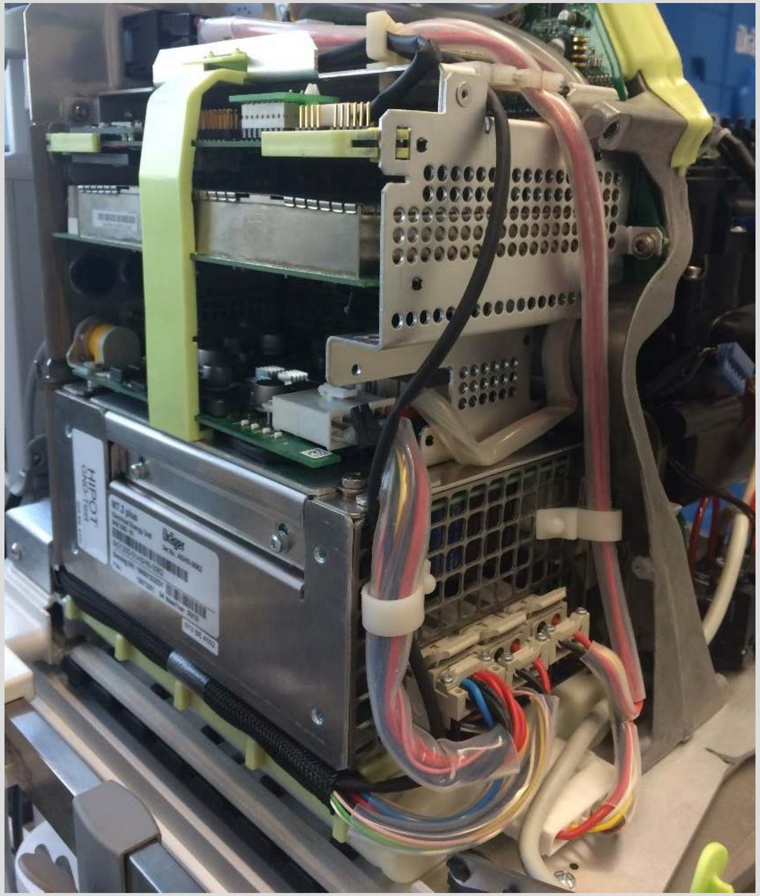
* operation time is shown as result of battery test every 3 months

new battery concept "nebaco" with power supply M7.3plus 8421200

(available from 01/2017 as conversion kit, see TSB 216)

- the internal battery of V-unit will be installed, also if the option PS500 installed too. (M7.3plus charges internal battery and PS500 battery).





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how to detect **nebaco** is installed:

nebaco with M7.3plus is **installed**

nebaco with M7.3plus is **<u>not</u> installed**

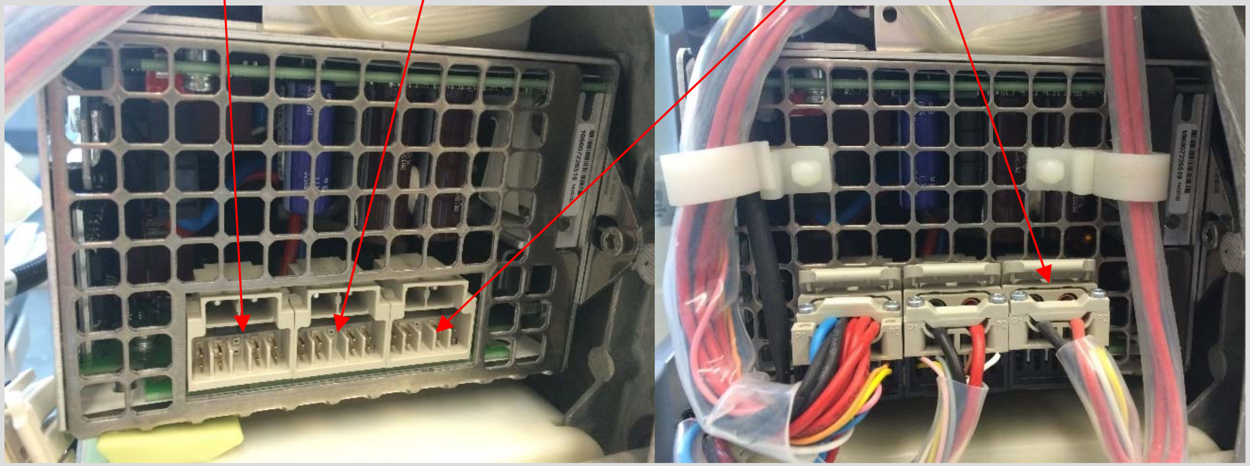
Workstation Critical Care/Neonatal Care; EvitaV300
comparison M7.3plus and M7.3classic

new: M7.3**plus** 8421200

24V-supply
V-unit

battery
PS500 (PB)

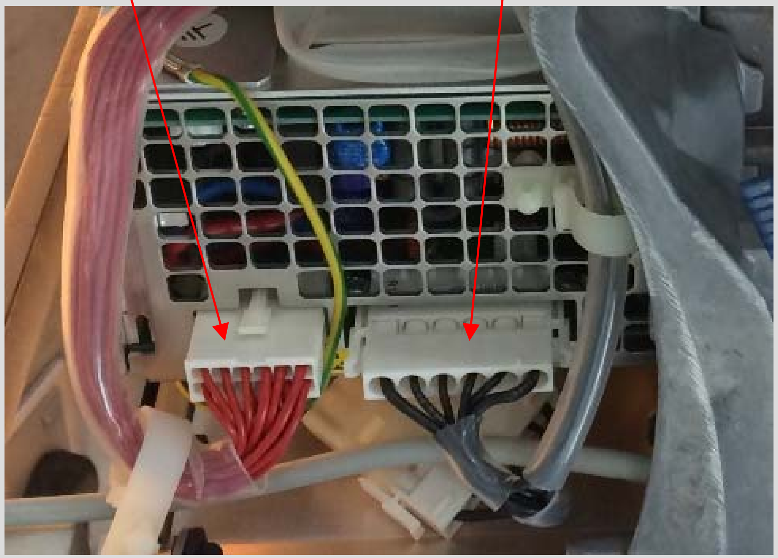
battery internal (NiMH)



M7.3**classic** 8415328

24V-supply
V-unit

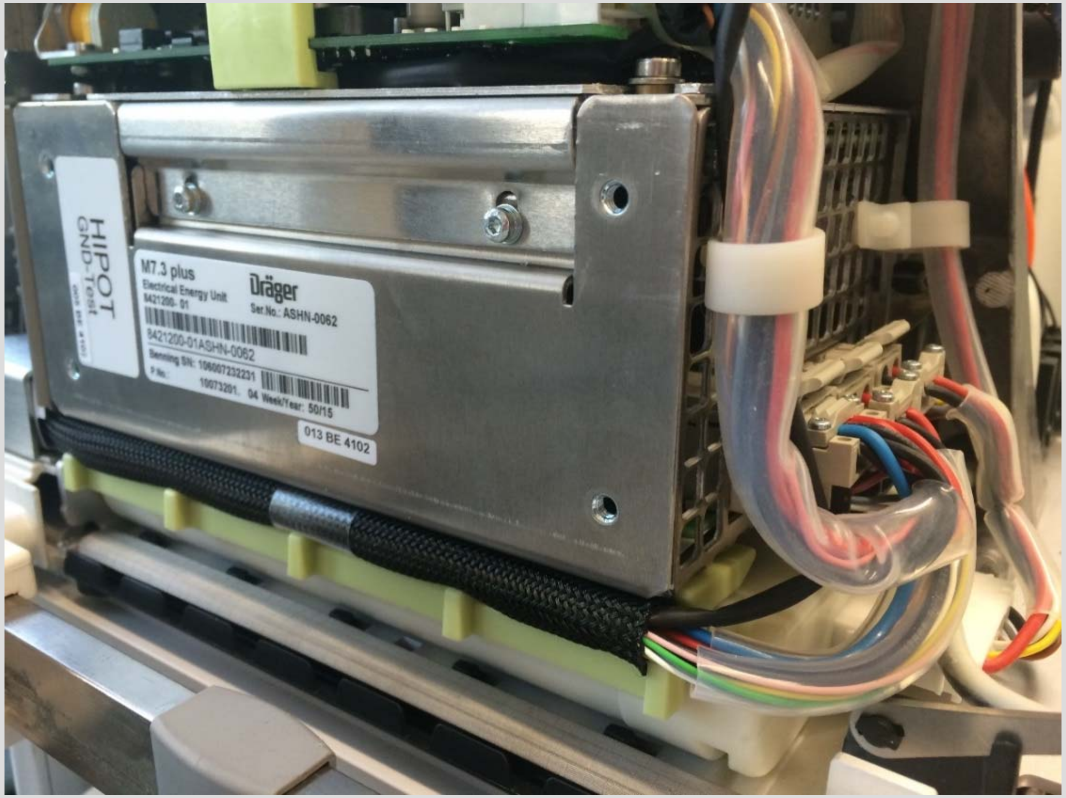
battery internal (NiMH)
or PS500 (PB)



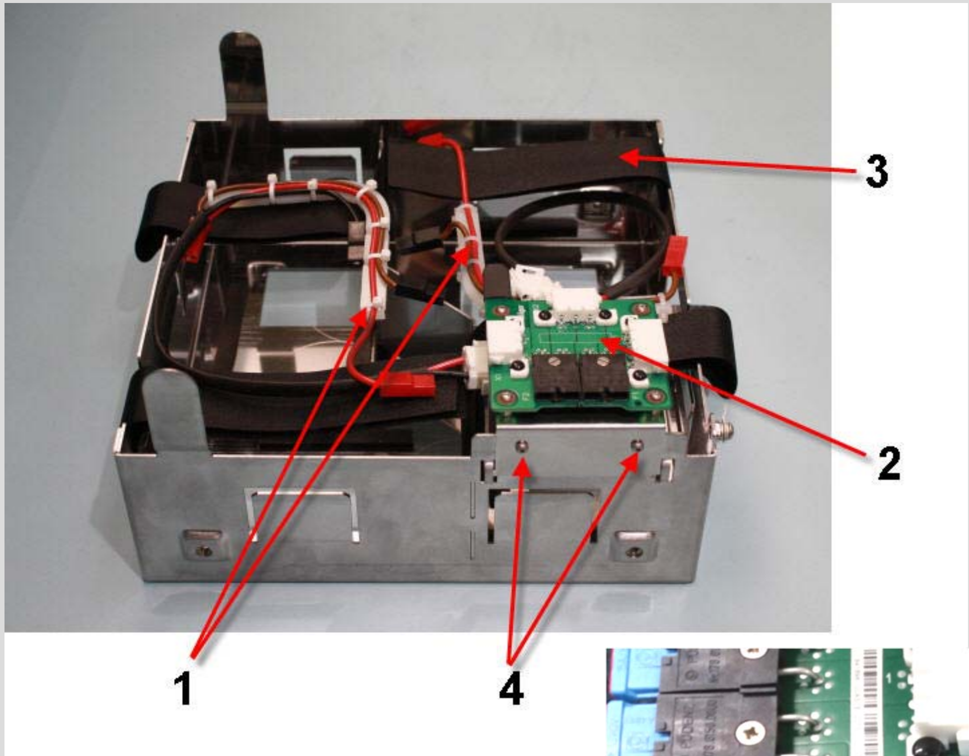
Workstation Critical Care/Neonatal Care; EvitaV300
nebaco with M7.3plus

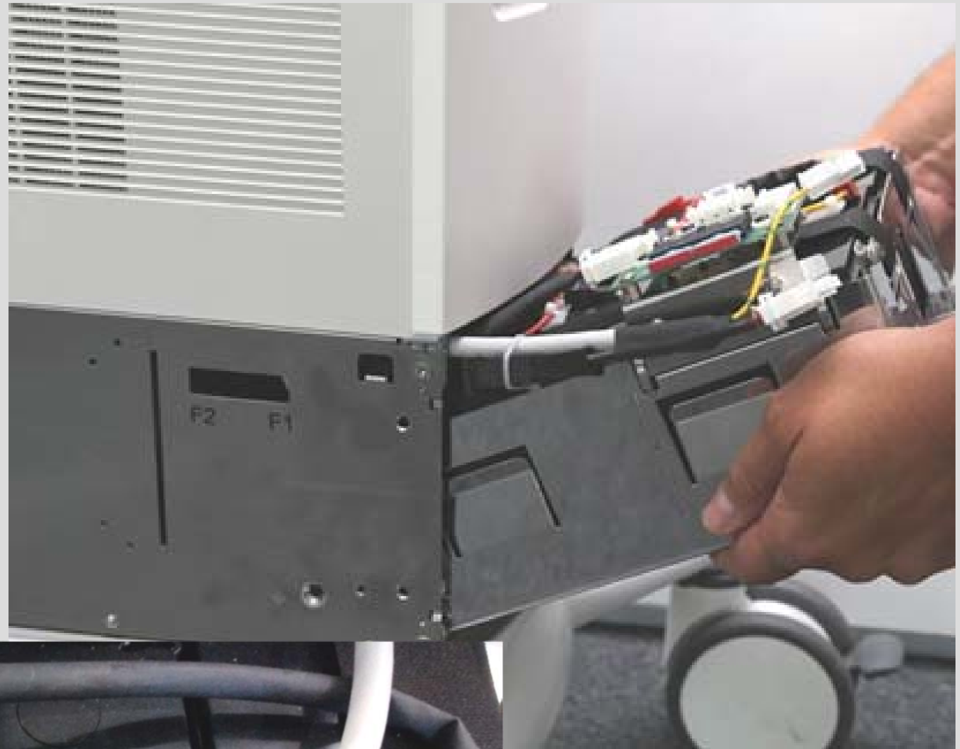
forecast: it´s **planned** to switch the **fabrication of V-units to M7.3plus in Q1/2017, depending on operational availability.**

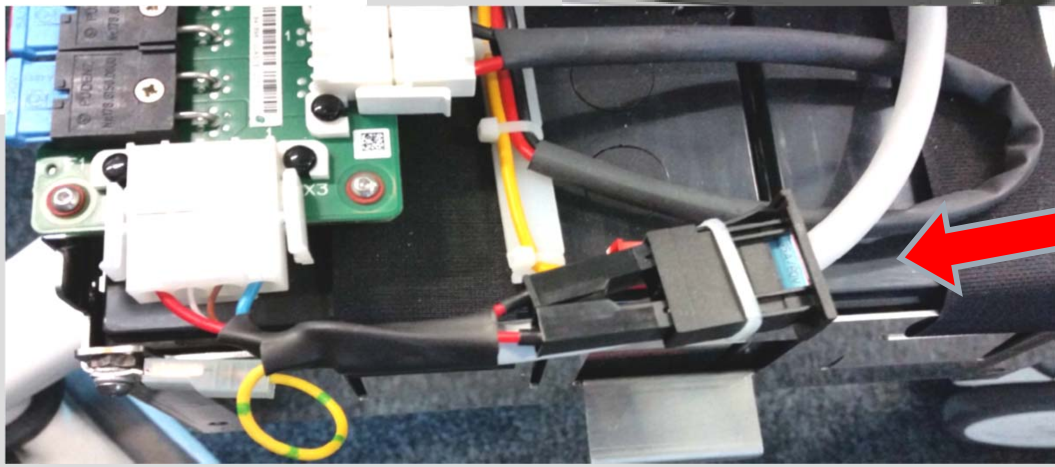
- this means from this day all produced V-units have a power supply M7.3plus and software 2.50 or higher, independent if option PS500 is installed or not.



Workstation Critical Care/Neonatal Care; EvitaV300
option PS500 with M7.3plus







pos.	designation
1	battery cable
2	PCB-battery box
3	hook and loop tape
4	mounting screws

This new, additional fuse in PS500 is limiting the current in the cable from M7.3plus to PS500.

Workstation Critical Care/Neonatal Care; EvitaV300
option PS500 with M7.3plus

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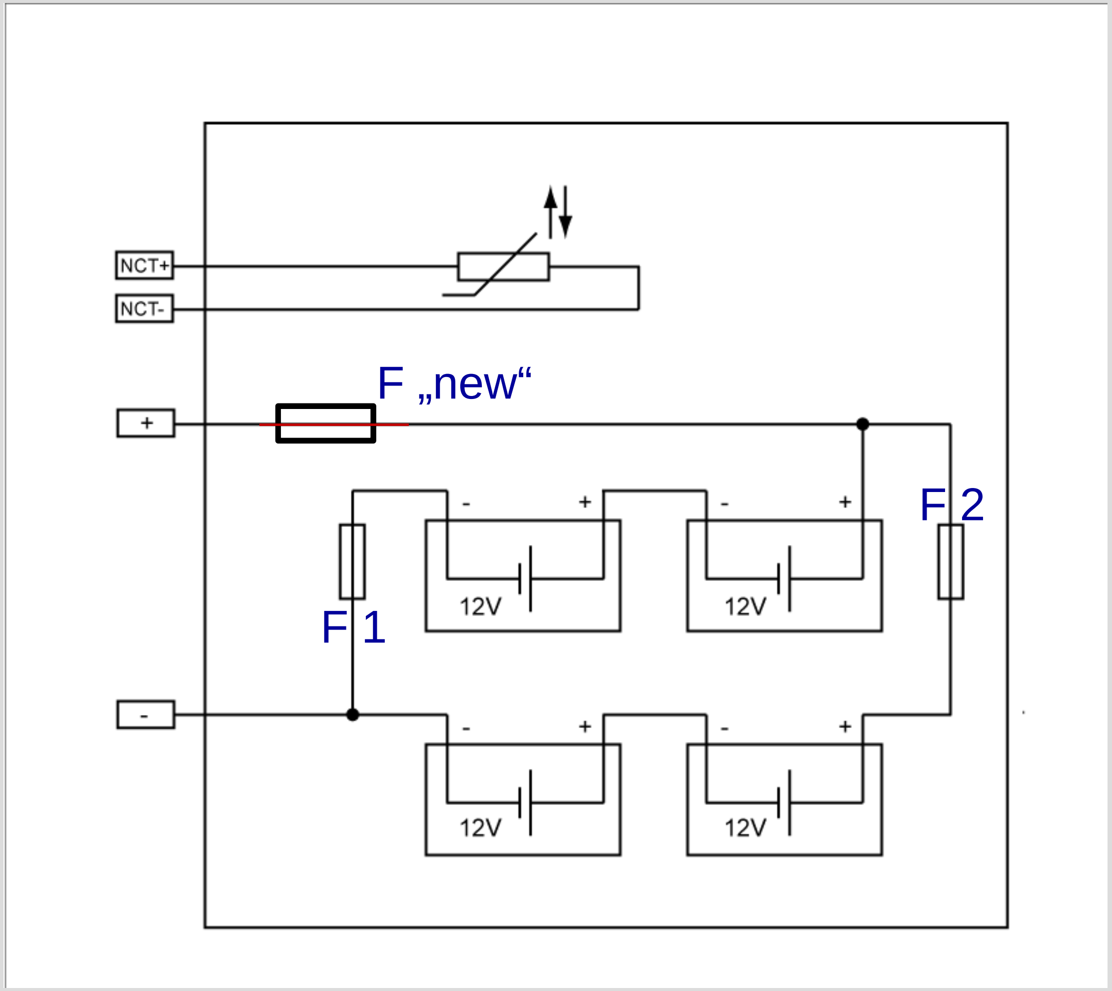
PS500 with M7.3plus

- In PS500 are 4 lead-batteries installed, charged direct from power pack M7.3plus.



F1 & F2 in PS500: limit the current of each battery-sting to 15 ampere

F3 in V-Unit with **M7.3classic**: F3 limit the current in battery cable to PS500 to 15 ampere

F3 in V-Unit with **M7.3plus**: F3 limit the current in battery cable to internal NiMH battery to 15 A.

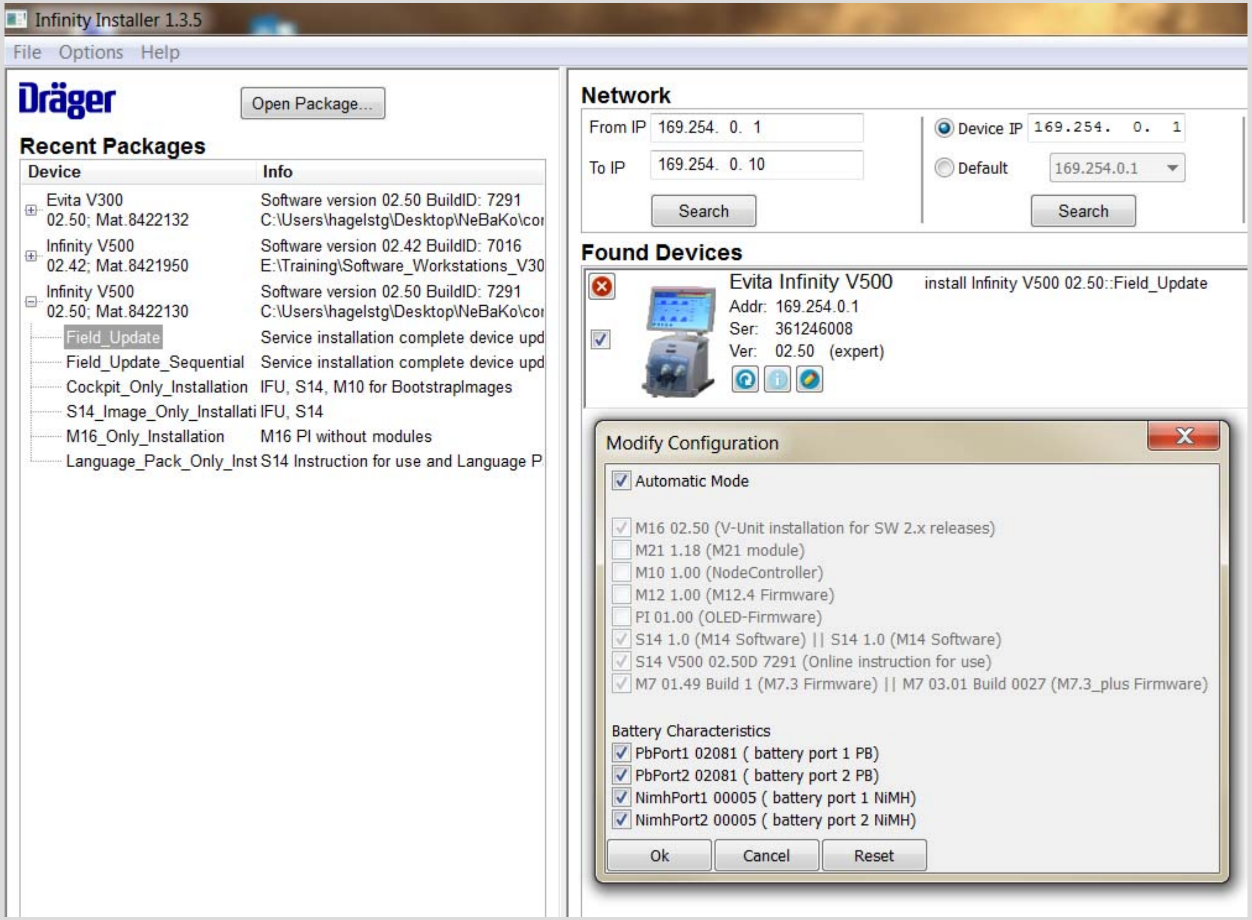
„**F new**“, the new fuse in PS500,
limit the current in battery cable to PS500 to 15 A.
If the V-Unit need to be „power-off“ pull F1+F2+F3 !

(F „new“ need to be pulled off in case of a failure.)

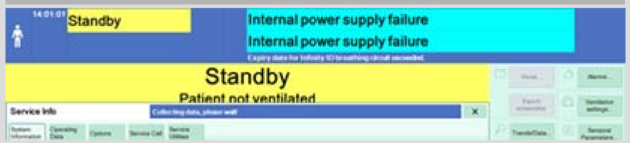
Workstation Critical Care/Neonatal Care; EvitaV300

nebaco with M7.3plus and new software version

For the **new battery concept** with power supply **M7.3plus** 8421200 the **new** product software **version 2.50** according conversion instruction need to be installed.



Note:
The power supply unit M7.3 plus (8421200) is unknown to device software versions prior to 2.50. During conversion according to conversion instructions 8422294 chapter 3 a device with software prior to 2.50 will issue an alarm "internal power supply failure" after installation of M7.3 plus and before installation of a device software 2.50 or higher, which can be ignored.



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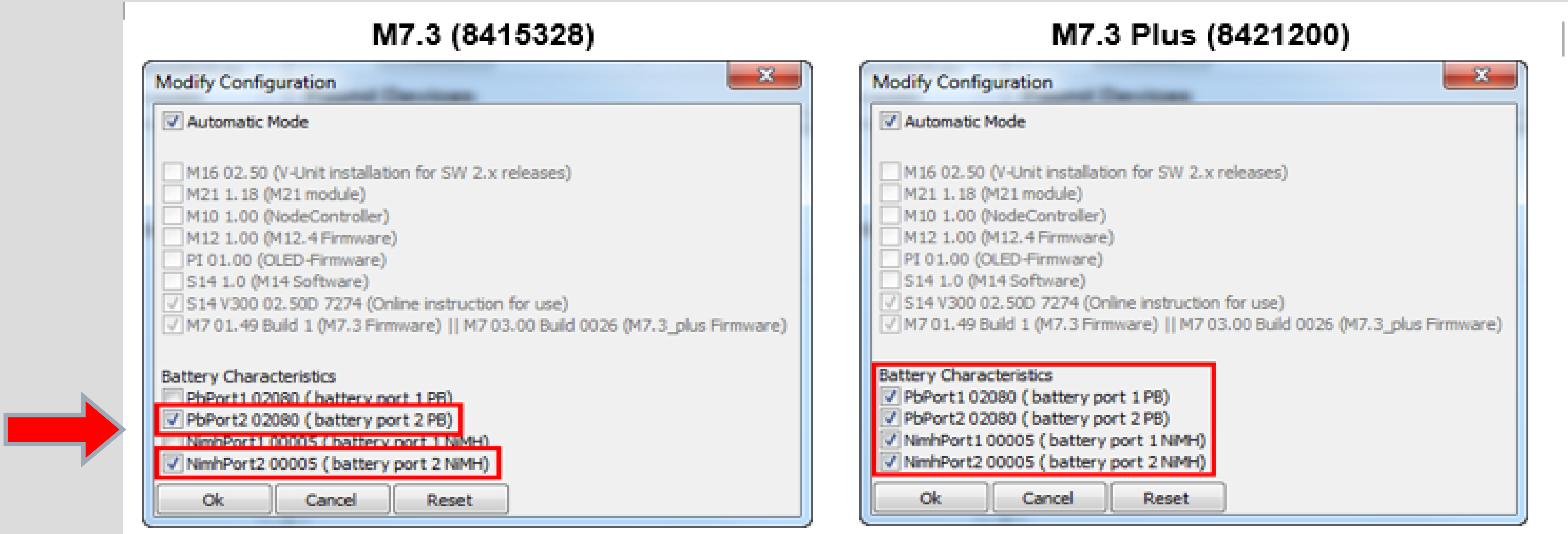
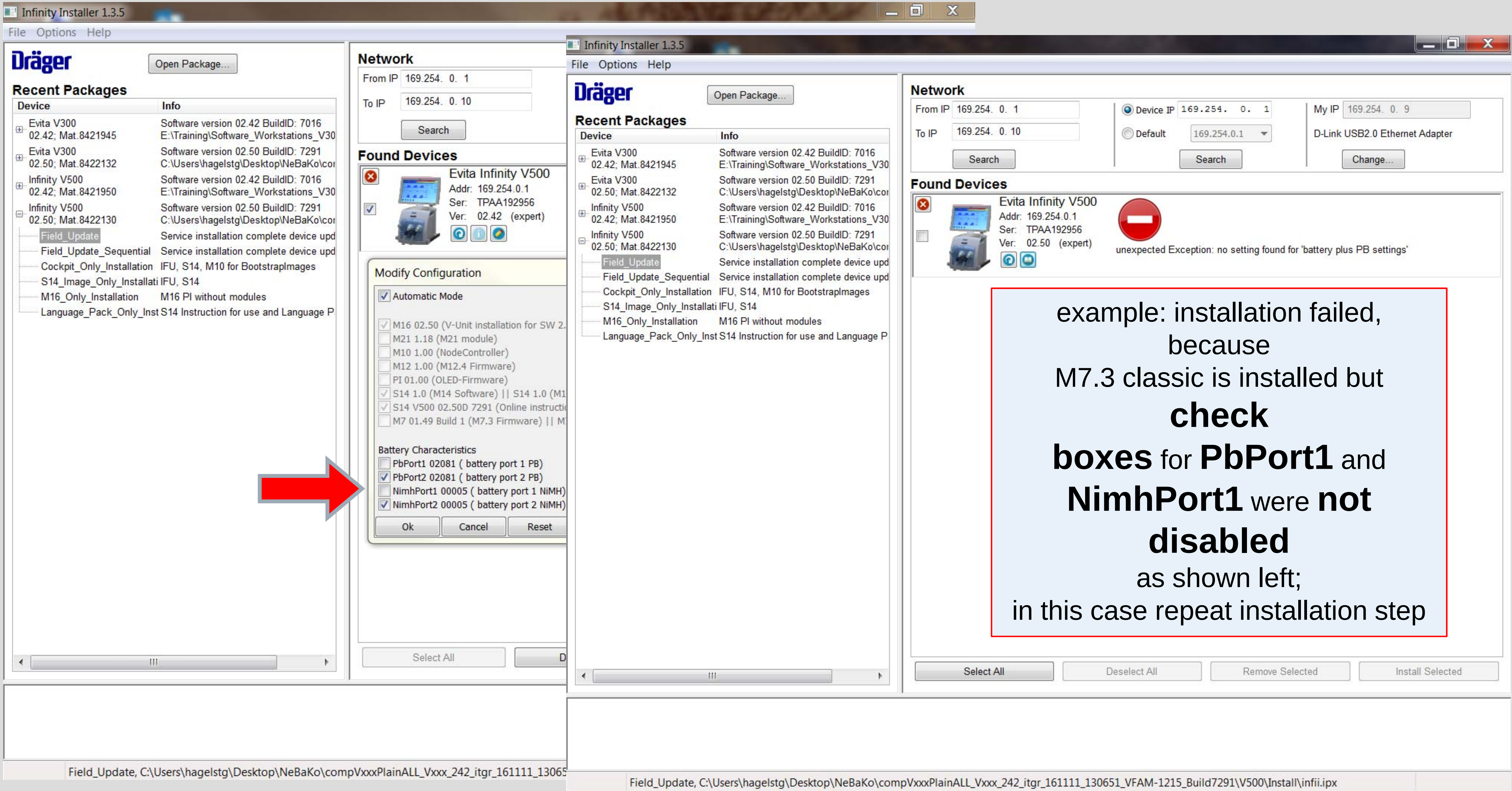
Note: **before** you **install** software **2.50** disable the check boxes for “PBPort 1” and “NimhPort1” in Infii, if **M7.3 “classic”** 8415328 is installed.

If you don`t do so, the software installation failed. (please **follow conversion instruction**!)



Workstation Critical Care/Neonatal Care; EvitaV300

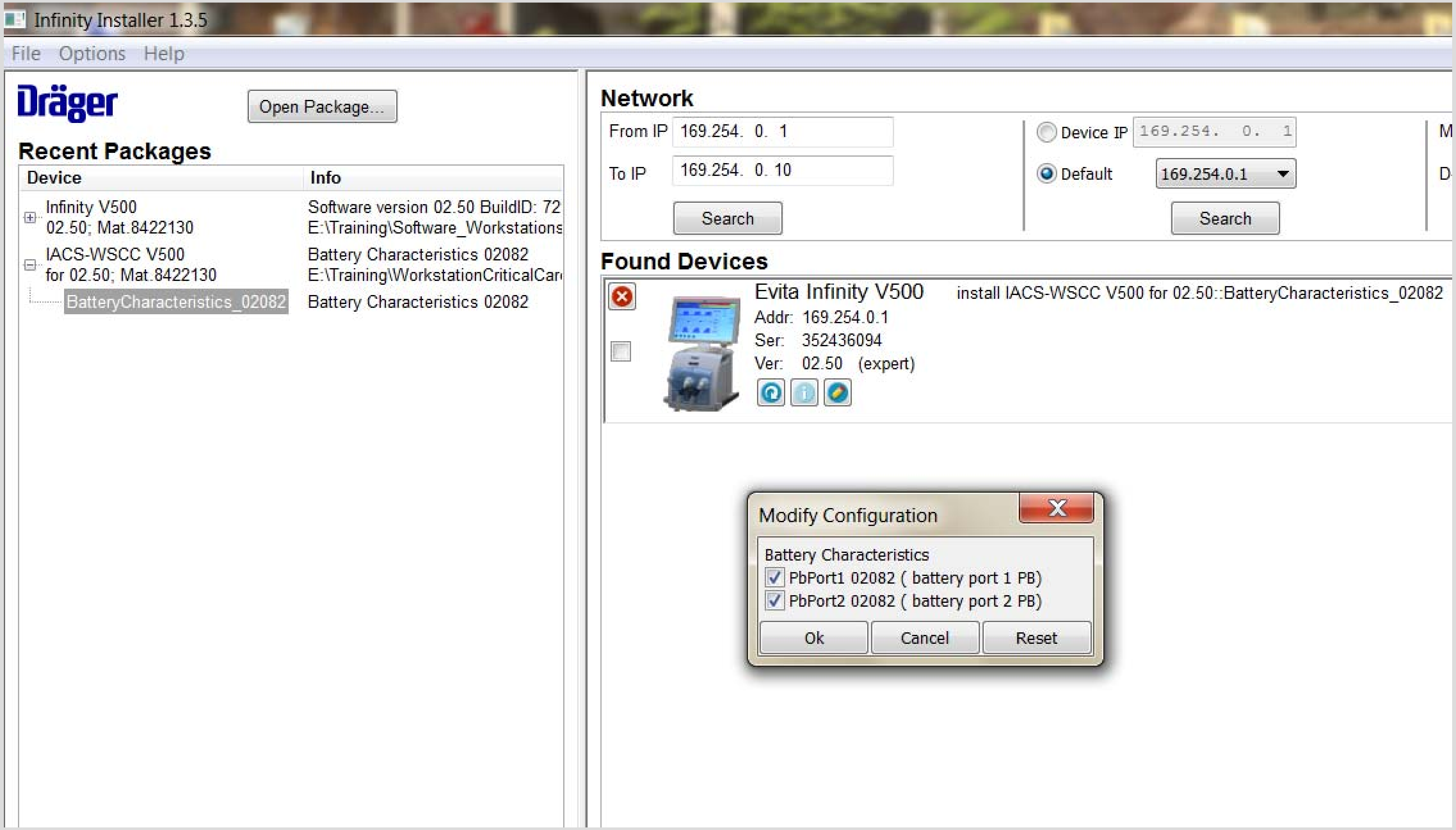
nebaco with M7.3classic – software installation



Workstation Critical Care/Neonatal Care; EvitaV300

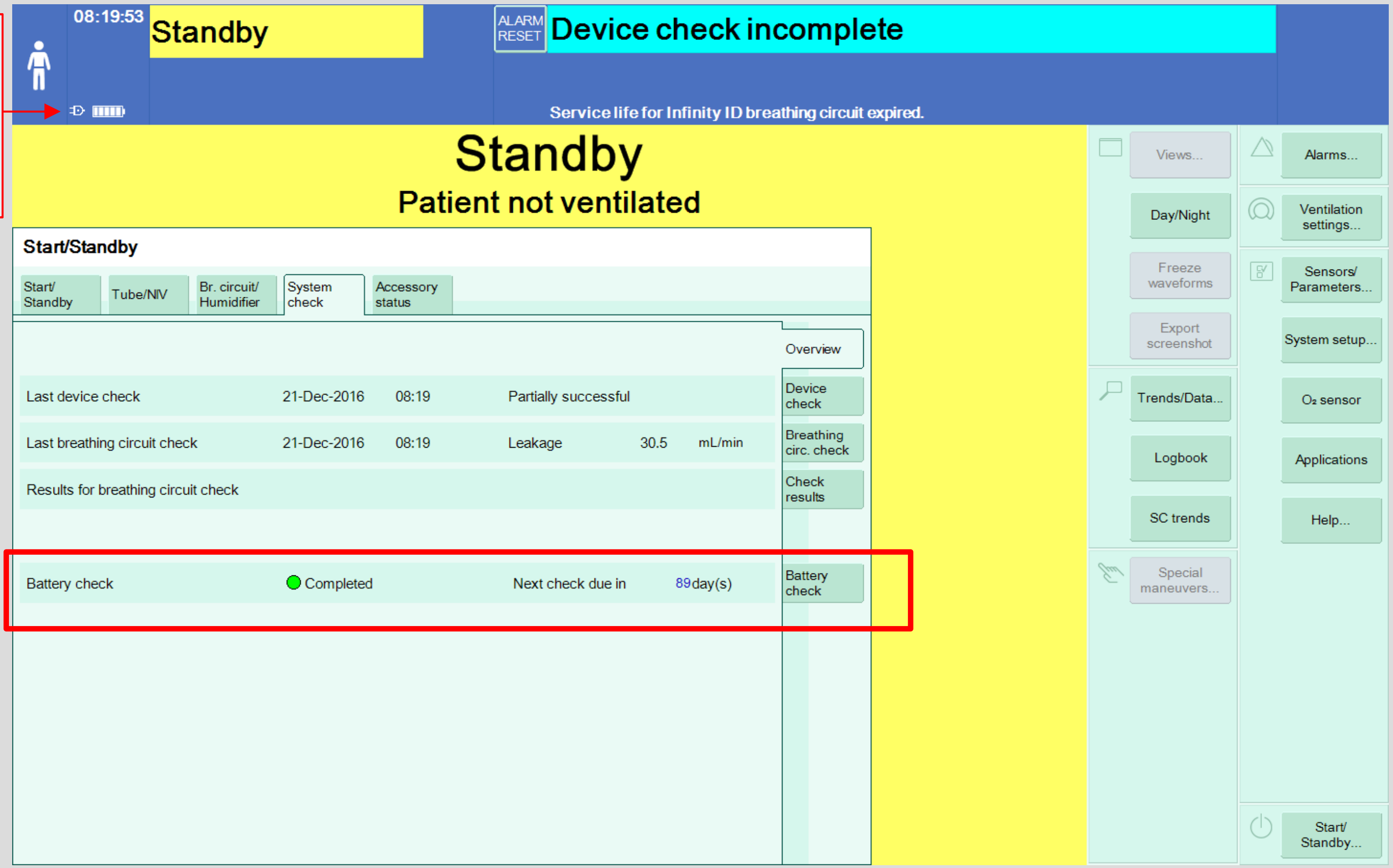
ne-ba-co with M7.3plus – software installation

According to **TSB 224** an additional battery characteristics 02082 must be installed on devices with new M7.3plus and software 2.50! In software 2.51 the BC 02082 is integrated.



V-units with **M7.3plus** will have with software 2.50 a **battery check** and a **new IfU** will be provided (independent if PS500 is installed or not).

battery symbol represents the charge of <u>all</u> installed batteries



Workstation Critical Care/Neonatal Care; EvitaV300

nebaco with M7.3plus – battery check

Preventive maintenance

WARNING
Risk due to defective components

Device failure is possible due to wear or material fatigue of the components. To maintain the function of all components, this device must be inspected and serviced at the intervals specified by the manufacturer.

Maintenance intervals

Component	Interval	Measure	Personnel responsible
Ambient air filter	Every 4 weeks	Cleaning, replace if necessary, see page 261	User
	Every 12 months	Replace, see page 261	User
Diaphragm of the expiratory valve	Every 12 months	Replace, see page 261	User
Expiratory valve	Every 2 years	Replace, see page 262	User
GS500: Breathing gas filter in the blower unit	Every 12 months	Replace, see page 262	Service personnel
GS500: Filter mat	Every 12 months	Replace, see page 262	Service personnel
Batteries	Every 3 months	Check capacity, see page 264	Service personnel
		Replace if necessary	Experts
	Every 2 years	Replace	Experts
Air filter (in the Air gas inlet)	Every 2 years	Replace	Experts
O_2 filter (in the O_2 gas inlet)	Every 6 years	Replace	Experts

The **new Instruction for Use** consider, that with the new battery concept a **battery check** should performed **every 3 months**.
This is necessary to check the **actual condition** of installed batteries!

Also provide the IfU **detailed information's** about new **battery concept** and **alarm behavior** (see next slides).

Workstation Critical Care/Neonatal Care; EvitaV300

nebaco with M7.3plus – battery check

Battery check

A battery check is required at regular intervals to determine the current state of the batteries. The battery check determines the approximate operating time.

The battery check consists of a charge-discharge-charge cycle. After the batteries have been fully charged, the device is operated in test mode with power supply from the batteries. The determined operating time is the approximate operating time to be expected in the next period of battery operation with typical ventilation without GS500.

Dräger recommends the following test intervals:

Internal battery (NiMH)	Every 3 months
PS500 power supply unit (VLRA)	Every 3 months

Prerequisites for the battery check

- The device is connected to the central gas supply.
- The device is connected to the mains power supply.
- The device is prepared and ready for use.
- The test lung is connected.
- A ventilation pattern is set, e.g.:
  - PC-AC
  - FiO_2 = 21 %
  - RR = 12/min
  - Pinsp = 20 mbar (or hPa or cmH_2O)
  - PEEP = 5 mbar (or hPa or cmH_2O)

The following table shows the typical operating time to be expected as a function of the ageing of a new battery.

If the batteries do not correspond to the approximate operating time listed, replacement of the batteries is recommended.

Age of the battery	Operating time of the internal battery (NiMH) with a full charge	Operating time of PS500 (VRLA) with a full charge
3 months	29 min	225 min
6 months	28 min	210 min
9 months	27 min	195 min
12 months	26 min	180 min
15 months	25 min	165 min
18 months	24 min	150 min
21 months	23 min	135 min
24 months	22 min	120 min

NOTE
The operating time may be reduced due to the utilization of the battery. The data are approximate values and cannot be regarded as guaranteed for every battery.

NOTE
Replace the batteries if the operating time falls below the minimum value (see chapter "Battery ageing" on page 351) or after 24 months.

Note (additional info G.H.):
If during 6 months a battery check was not performed,
the user will be informed by an alarm message.
This alarm can be reset/confirmed by the user. 01/2017 *G.H.*

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Battery concept

General information

At the time of manufacture and delivery, batteries have a typical capacity which is in accordance with the information specified in the battery manufacturer's data sheet. The electrochemical composition of the battery is the determining factor for its total capacity. The operating time of the batteries derived from these specifications can be found in the "Technical Data" chapter.

NOTE
The capacity of batteries changes with increasing age and utilization.

All the following information and specifications refer to perfectly functioning batteries. If batteries are defective or faulty, the functional integrity, e.g., battery charge indication or alarms, may be impaired. See chapter "Battery check" on page 264.

Display of battery charge

The battery charge indication shows the available battery charge determined by the electrochemical processes. When the batteries are, e.g., fully charged, this state is indicated by a corresponding symbol.

Symbol	Battery charge
▮▮▮▮▮	90 to 100 %
▮▮▮▮	60 to <90 %
▮▮▮	40 to <60 %
▮▮	20 to <40 %
▮	<20 %, flashes light and dark red in 1-second pulses.
▯	Batteries defective or no information available on the battery charge.

The battery charge indication is a relative indication which is based on the electrochemical properties of the battery. The battery charge indication is evaluated on the basis of a battery model.

The use of a model-based indication is a state-of-the-art technique which finds application in many fields, e.g., computers, mobile phones, etc.

The model-based indication of battery charge takes account of the following information, among other things:
– Type of battery (e.g., NiMH or VRLA)
– Maximum capacity on delivery (e.g., 12 Ah)
– Age of the battery (e.g., new or 2 years)
– Capacity spent (irreversibly lost) over the utilization time (e.g., 1000 Ah)
– Present power requirement of the device (power consumption, e.g., 2.5 A)
– Discharge mode
– Charging mode

If the power consumption changes, e.g., due to switching to Standby, operation of a GS500, or adjustment of the screen brightness, the remaining available operating time of the device also changes. The battery charge indication is updated to take account of the present power requirement (power consumption).

In accordance with the specification, the battery charge indication is only displayed and updated after the device has been completely started up. This procedure may take a few minutes.

Battery ageing

The electrochemical composition of a battery alters as a result of ageing and utilization. Consequently every battery loses a proportion of its maximum capacity in comparison with its new condition. This loss of capacity is typically irreversible.

As a result of ageing and utilization of the battery, there is a change in the actual maximum operating time which is displayed by the percentage values in the battery charge indication. The percentage value refers to the currently available operating time of batteries which were fully charged before use.

New batteries

The following data for minimum operating time apply to new and fully charged batteries. The ▮▮▮▮▮ symbol for a fully charged battery is displayed. See also the "Display of battery charge" chapter and the "Technical Data" chapter. Owing to production fluctuations during the manufacture of batteries, the operating time can be considerably longer.

Battery used (Battery type)	Minimum operating time without operation of a GS500	Minimum operating time with operation of a GS500
Internal battery	30 min	15 min
PS500	240 min	120 min

Aged batteries, e.g., 2 years old

The following data for minimum operating time apply, e.g., to 2-year old and fully charged batteries. The data are approximate values and cannot be regarded as guaranteed for every battery, The ▮▮▮▮▮ symbol for a fully charged battery is displayed. See also the "Display of battery charge" chapter and the "Technical Data" chapter.

Battery used (Battery type)	Minimum operating time without operation of a GS500	Minimum operating time with operation of a GS500
Internal battery	22 min	11 min
PS500	120 min	60 min

NOTE
A reduction in the capacity of batteries due to ageing and utilization is normal. As a greatly simplified approximation, an average linear reduction in capacity can be assumed. The current individual capacity of a battery depends on the following factors, among others:
– Age
– Utilization (frequency, duration, and power consumption)
– Battery charge
– Ambient temperature

Spent batteries

NOTE
When the internal battery and the batteries in the PS500 show less than the following residual operating times, they are considered to be spent.

Battery used (Battery type)	Spent with remaining residual operating time without operation of a GS500
Internal battery	<22 min
PS500	<120 min

Workstation Critical Care/Neonatal Care; EvitaV300

nebaco with M7.3plus – battery check

Alarm behavior in battery operation

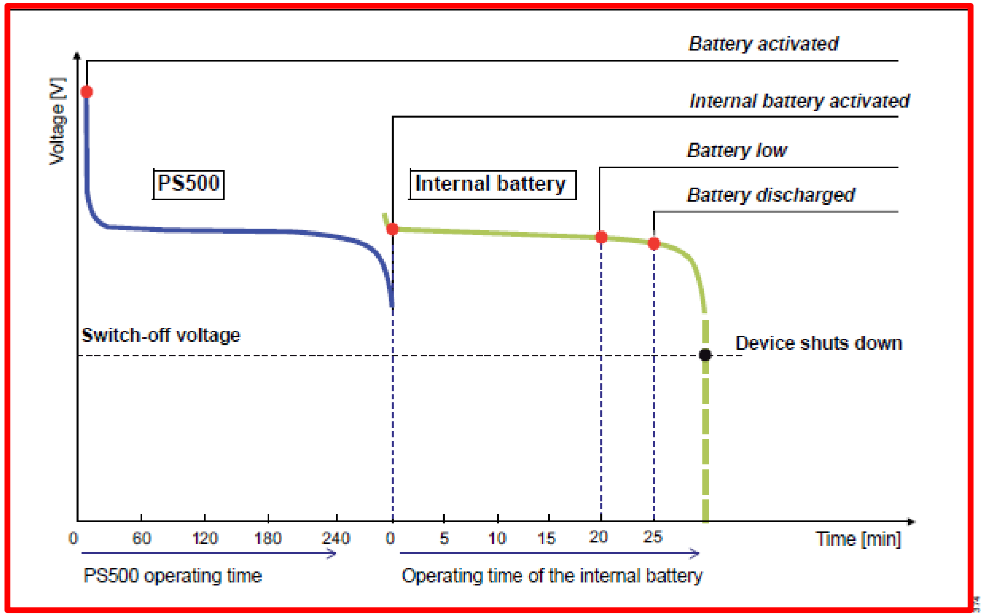
The switch-over of the device to battery operation is indicated by the ***Battery activated*** alarm (see chapter "Alarm – Cause – Remedy"). The alarm can be acknowledged by the user. Consequently the ***Battery activated*** alarm will no longer be displayed until the mains power supply is re-established.

When the device is equipped with a PS500 power supply unit, in battery operation the PS500 is discharged first. If the mains power supply has not been re-established, the device switches over to the internal battery after the operating time of the PS500 has elapsed. The switch-over is indicated by the ***Internal battery activated*** alarm.

At the end of the operating time of the internal battery, the device generates the ***Battery low*** alarm. The ***Battery discharged*** alarm follows after that.

These alarms appear at the time specified by the model-based calculation for the particular battery.

Schematic representation of the sequence of alarms



The schematic representation of the sequence of alarms with respect to battery utilization is shown in an example with a PS500 but without the use of a GS500. The representation corresponds to the operating time of new and fully charged batteries.

When the voltage drop of the internal battery reaches an operationally critical value, a shut-down of the device due to an inadequate supply is immediately imminent. In this case the power supply failure alarm sounds.

NOTE
If the device displays the ***Battery low*** alarm message or the ***Battery discharged*** alarm message, connect the device to the mains power supply.

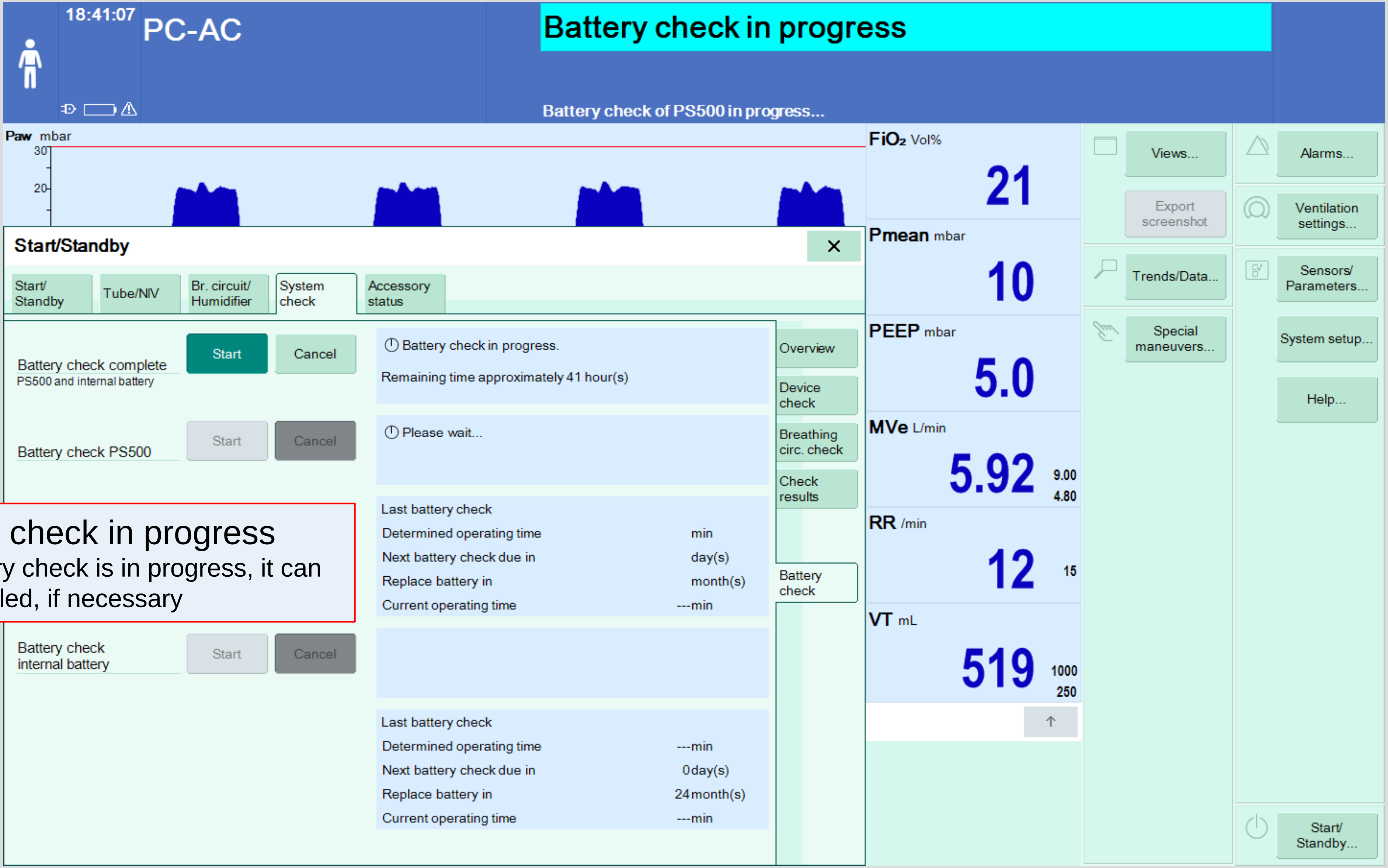
NOTE
When the remaining calculated operating time is less than 10 minutes, the model-based ***Battery low*** alarm appears. When the remaining calculated operating time is less than 5 minutes, the model-based ***Battery discharged*** alarm appears.

NOTE
The operating time remaining after the corresponding alarms can be considerably longer than the specified minimum operating time.

NOTE
When the device is fitted with the GS500 gas supply unit, the device calculates the time for the ***Battery discharged*** alarm allowing for the power consumption of a GS500, regardless of whether the GS500 is activated or not.

Workstation Critical Care/Neonatal Care; EvitaV300

nebaco with M7.3plus – battery check



battery check in progress

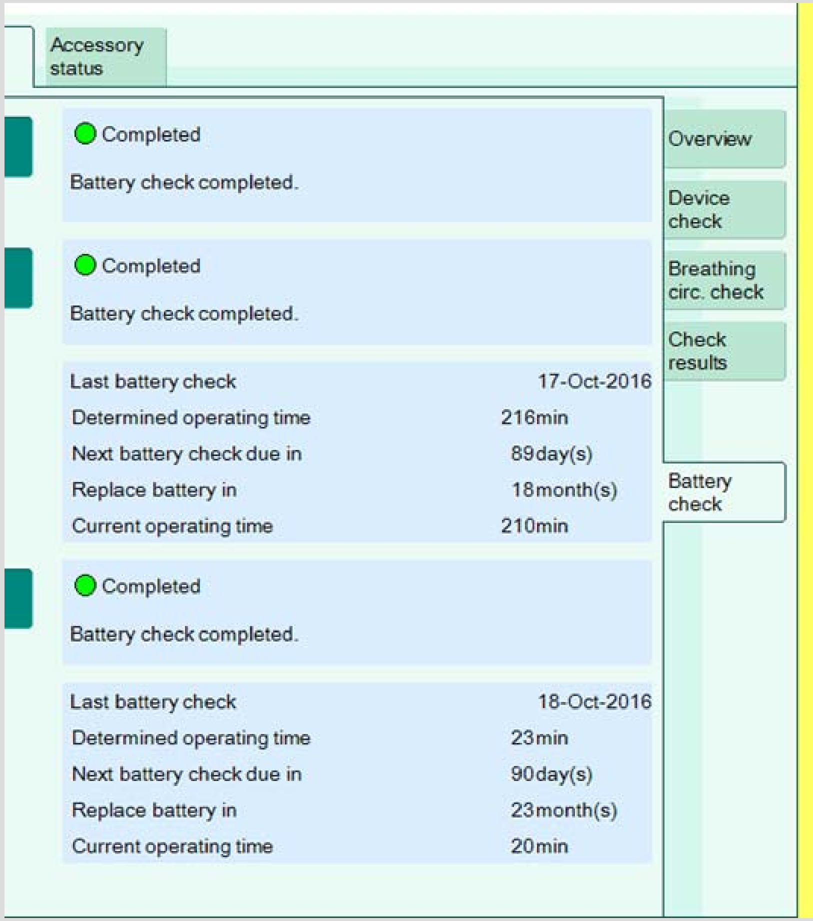
if a battery check is in progress, it can be canceled, if necessary

Workstation Critical Care/Neonatal Care; EvitaV300
nebaco with M7.3plus – battery check

Battery check in correct order (overview):

1. charge complete the internal NiMH battery
2. charge complete the external VLRA battery (PS500)
3. discharge complete the external VLRA battery (PS500)
4. charge complete the external VLRA battery (PS500)
5. discharge complete the internal NiMH battery
6. charge complete the internal NiMH battery
7. when battery check is finished,

the **actual capacity** and the **month of next replacement** will be shown



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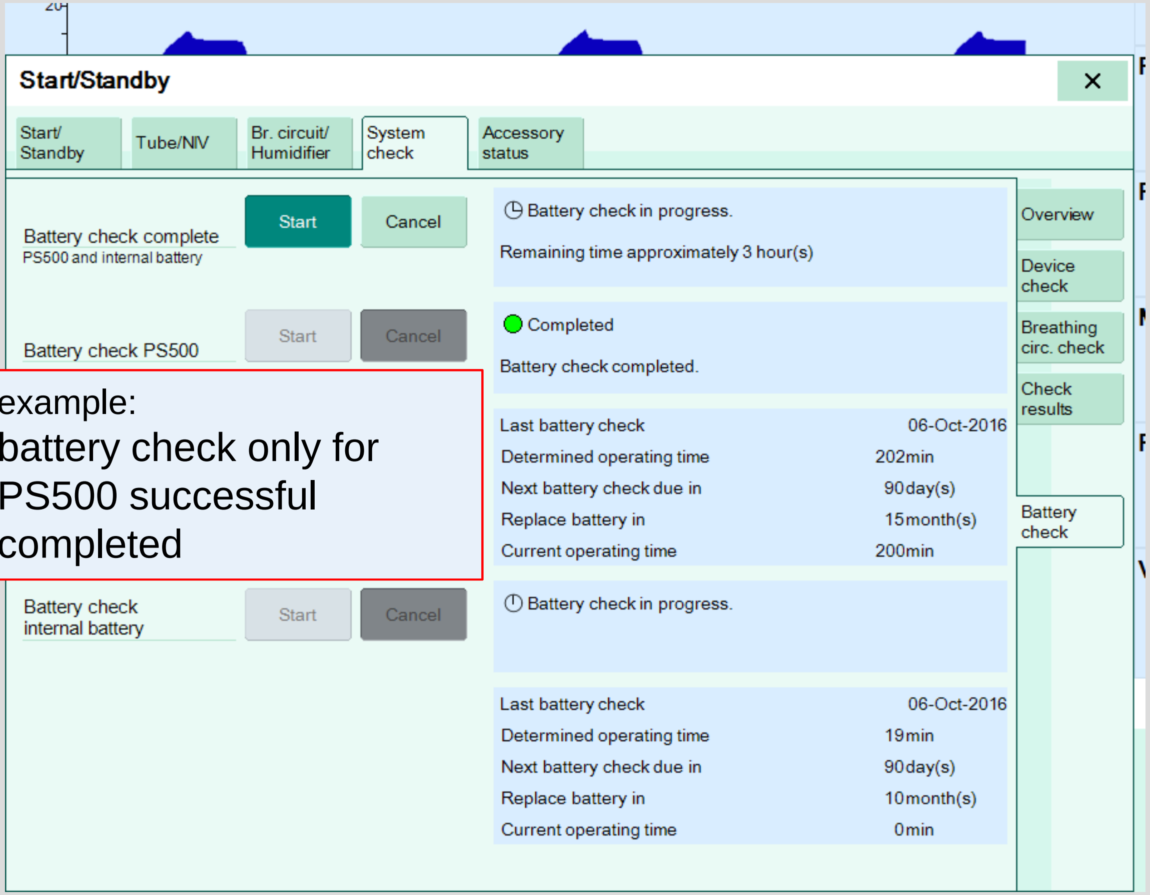
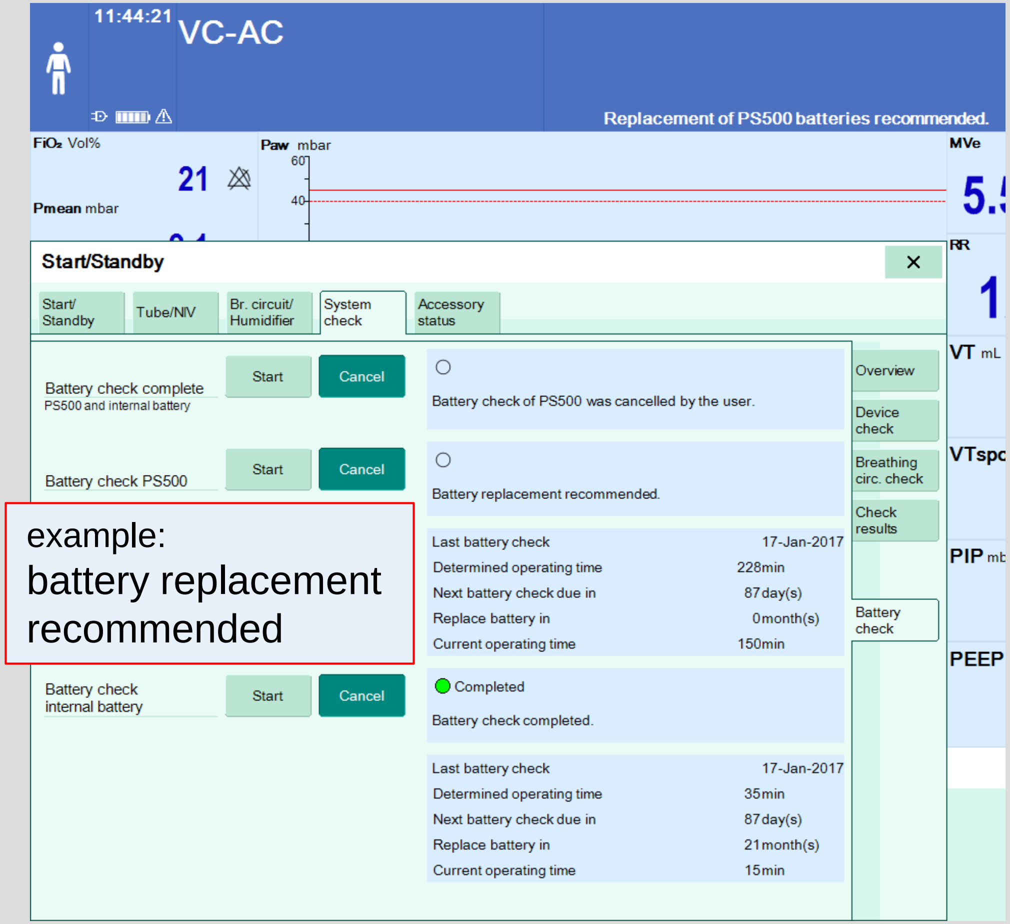
Workstation Critical Care/Neonatal Care; EvitaV300

nebaco with M7.3plus – battery check



example:
battery check only for PS500 successful completed



example:
battery replacement recommended

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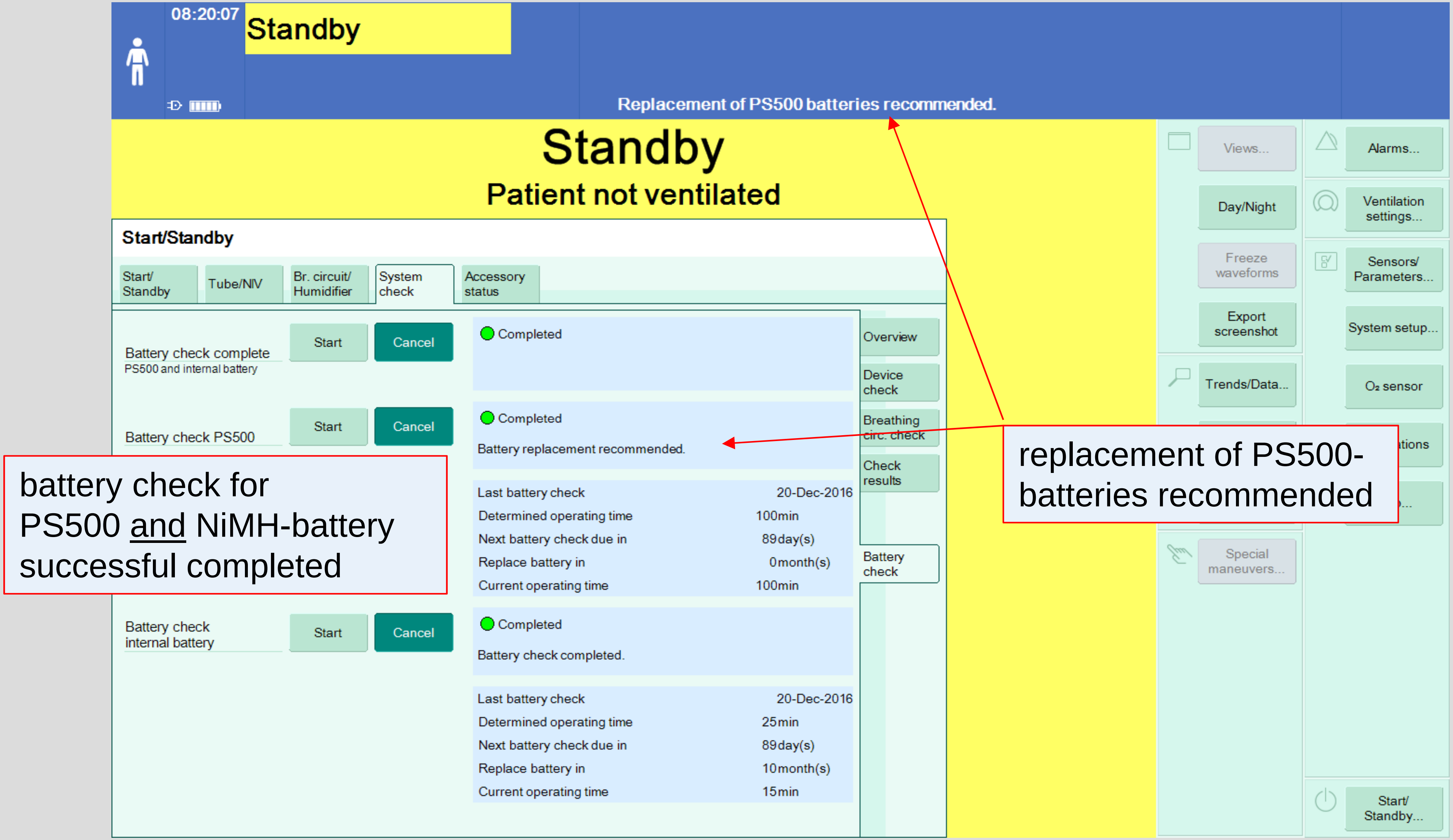
Workstation Critical Care/Neonatal Care; EvitaV300

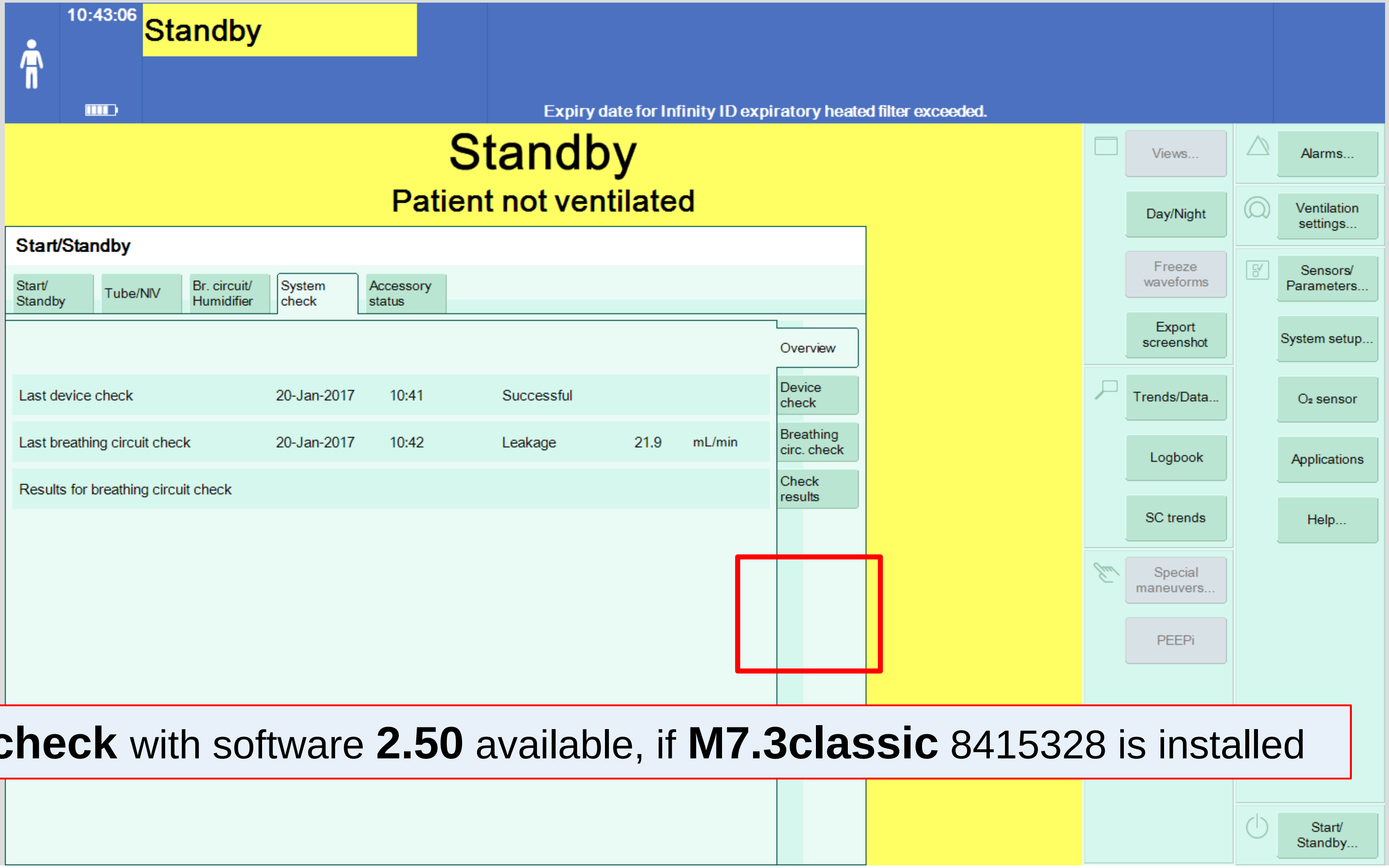
nebaco with M7.3plus – battery check



Workstation Critical Care/Neonatal Care; EvitaV300

V-Unit with M7.3 classic – no battery check

Note: **V-units** with **M7.3 classic** do **no**t have the **battery check** with software version **2.50,** neither a new IfU is provided, the **old IfU** is still **valid**.

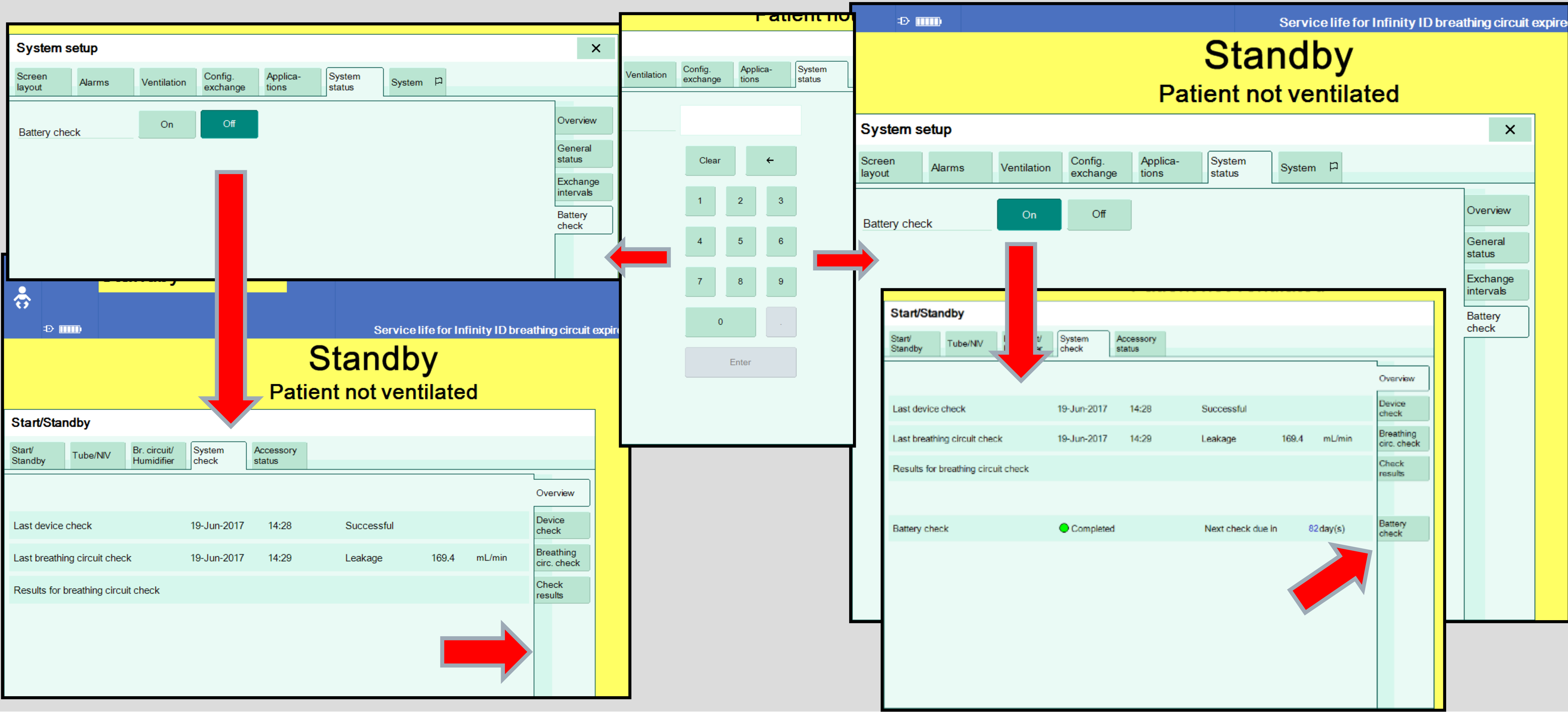


no battery check with software **2.50** available, if **M7.3classic** 8415328 is installed

Workstation Critical Care/Neonatal Care; EvitaV300

V-Unit with M7.3plus – no battery check

With software version 2.51 (planned for July 2017) it is possible, to disable the battery check on devices with M7.3plus.



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Note (repair information): M7.3 **Plus** (8421200):

If you **replace** a faulty power supply **M7.3plus** by a new M7.3plus, it`s **not necessary** to **replace** the internal **NiMH-battery** and either the **VLRA-batteries** from PS500. After installation of a new M7.3plus the **battery check** must be performed, to determine the actual **condition** of **batteries** and store this information in the new power supply. The service documentation will be adapted soon.

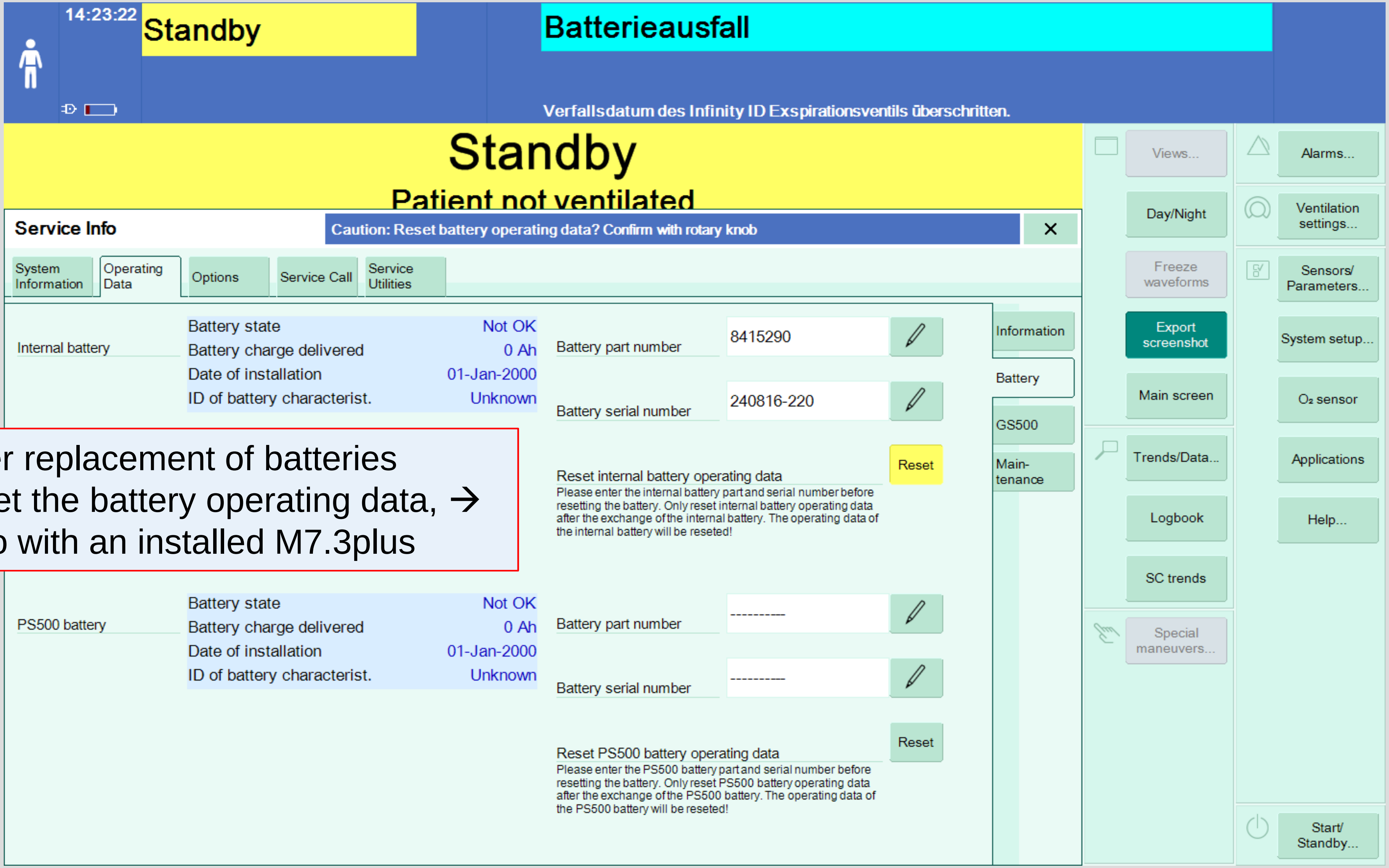
If you **replace** an **M7.3 classic** (8415328) with an **M7.3classic** (8415328) it`s still valid that you also have to **replace all installed batteries** by new ones and reset the battery operating data as repair instruction.

> **1 Replacing the power supply unit M7.3**
>
> **NOTE**
>
> When replacing a faulty power supply unit M7.3, the internal NIMH battery or the lead-gel batteries of the PS500 must also be replaced, as battery wear data saved in the power supply unit are no longer available. Otherwise the displayed available battery charge may be incorrect.

On the other hand if you **replace** in future a **M7.3classic** (8415328) with a **M7.3plus** (8421200), you need to hand over the **new instruction for use** etc., because battery check and other functionality is now available with M7.3plus.

Workstation Critical Care/Neonatal Care; EvitaV300

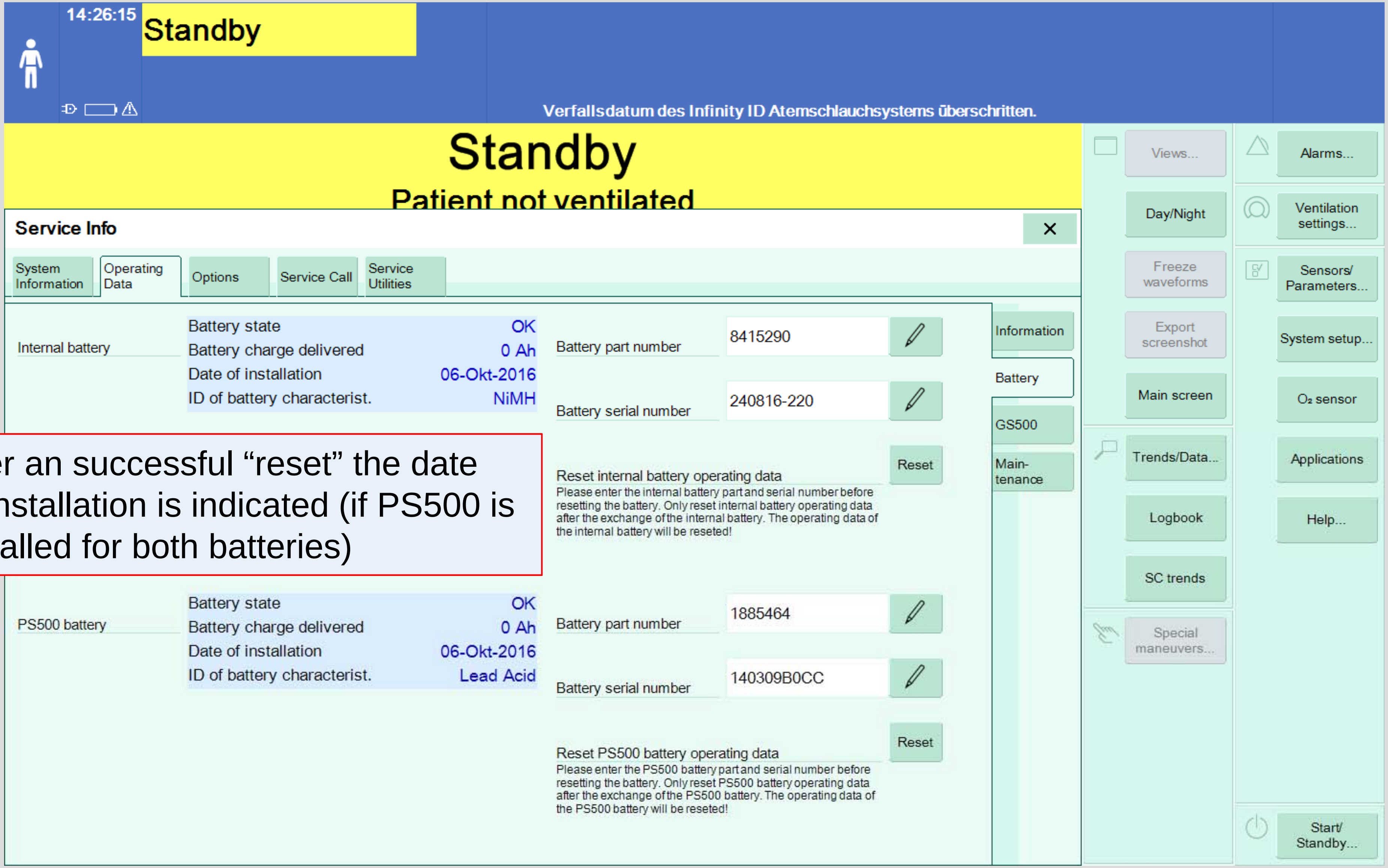
nebaco with M7.3plus – reset battery operating data



after replacement of batteries reset the battery operating data, → also with an installed M7.3plus

Workstation Critical Care/Neonatal Care; EvitaV300

nebaco with M7.3plus – reset battery operating data



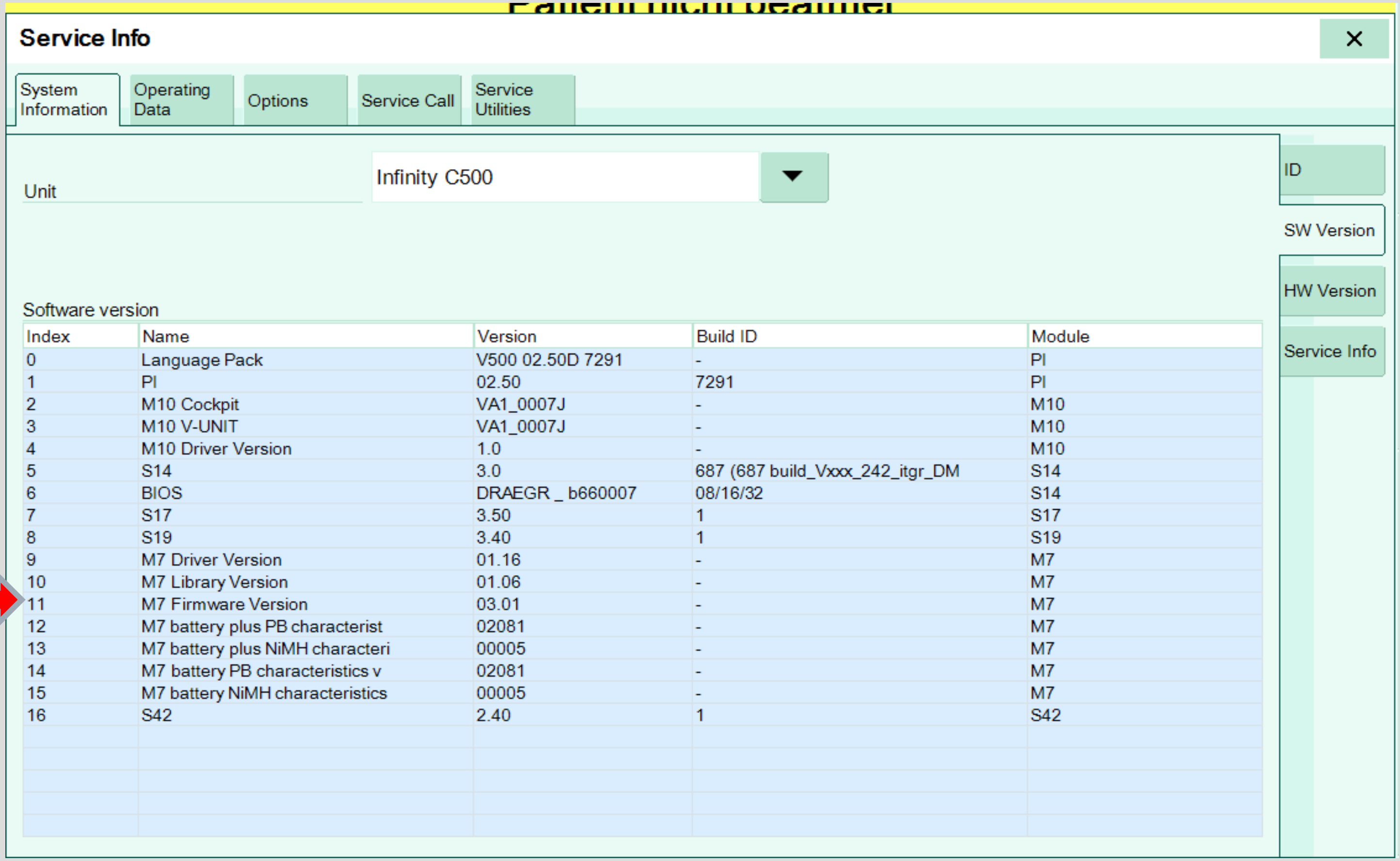
after an successful “reset” the date of installation is indicated (if PS500 is installed for both batteries)

Workstation Critical Care/Neonatal Care; EvitaV300

nebaco with M7.3plus – software & characteristics

Power supply **M7.3plus** needs a new **firmware** and **characteristic**

for NiMH-battery and VLRA-batteries (PS500)“. example below shows software 2.50:



Index	Name	Version	Build ID	Module
0	Language Pack	V500 02.50D 7291	-	PI
1	PI	02.50	7291	PI
2	M10 Cockpit	VA1_0007J	-	M10
3	M10 V-UNIT	VA1_0007J	-	M10
4	M10 Driver Version	1.0	-	M10
5	S14	3.0	687 (687 build_Vxxx_242_itgr_DM	S14
6	BIOS	DRAEGR _ b660007	08/16/32	S14
7	S17	3.50	1	S17
8	S19	3.40	1	S19
9	M7 Driver Version	01.16	-	M7
10	M7 Library Version	01.06	-	M7
11	M7 Firmware Version	03.01	-	M7
12	M7 battery plus PB characterist	02081	-	M7
13	M7 battery plus NiMH characteri	00005	-	M7
14	M7 battery PB characteristics v	02081	-	M7
15	M7 battery NiMH characteristics	00005	-	M7
16	S42	2.40	1	S42

Workstation Critical Care/Neonatal Care; EvitaV300

software overview w/o & with option PS500**

software version WSCC/ WSNC/ EvitaV300	Workstation Critical Care / Workstation NeonatalCare / EvitaV300 **w/o option PS500**				Workstation Critical Care / Workstation Neonatal Care / EvitaV300 **with option PS500***				
	power supply **firmware** M7.3	M7.3plus	**characteristic** NiMH battery	characteristic VLRA battery	power supply M7.3classic: firmware (fw)	characteristic NiMH-battery: 1	2	characteristic VLRA-battery: 1	2
2.31	fw version 1.41	---	00003	01034	not valid				
2.40	not valid				not valid				
2.41	fw version 1.43	---	00004	01035	fw version **1.49***	00004	---	01052	---
2.42	fw version 1.49	---	00004	01052	fw version **1.49***	00004	---	01052	---
2.50	M7.3 fw 1.49	**M7.3plus** fw 03.01	00005	02082	only with **M7.3plus*** and firmware 03.01	00005	00005	02082	02082

* in all V-units with option PS500 the new battery concept with power supply M7.3plus and software version 2.50 or higher must be installed according TSB 206.

** see also documentation in service connect

Note to TSB **220** „upgrade kit“ 2017-**01**-**05, class 2**

Countries Affected:	**USA, Australia, Bolivia** only
Concerned Product:	The following upgrade kits:
	1. Upgrade kit, 8416200, Critical Care Workstation with **Evita Infinity V500**
	2. Upgrade kit, 8419200, Neonatal Care Workstation with **Babylog VN500**
	3. Upgrade kit, 8420420, **Evita V300**
Short Description:	Reason / Solution: Potential malfunction / exchange of power supply modules 8421200 Rev.02 against **new version 8421200 Rev. 03**
	Replacement of power supply modules 8421200 Rev.02 as part of the upgrade kits.

TSB shall be completed until 2017-02-03

Note to TSB 216 „new battery concept“ 2017-01-18 class 1

Countries Affected: all (see list)

Concerned Product: Critical Care Workstation with **Evita Infinity V500**

Neonatal Care Workstation with **Babylog VN500**

Evita V300

Short Description: PS500 battery supply with Panasonic lead-batteries and additional NiMH-battery in a new battery concept with new software 2.50.

Target Due Date: **TSB shall be completed until 2017-10-31**

TSB 216 replaces TSB 206.

Dräger

Note to TSB **224** „new battery characteristic 02082 when use M7.3plus with software 2.50“ 2017-**03-21 class 3**

Countries Affected:	all (see list)
Concerned Product:	Critical Care Workstation with **Evita Infinity V500**
	Neonatal Care Workstation with **Babylog VN500**
	Evita V300
Short Description:	characteristic 02082 will provide more stability on devices with PS500 and new battery concept with software 2.50.
Target Due Date:	not applicable, but **TSB 224 shall be completed together with TSB 216**